

Russian spacecraft and will serve as an extra airlock for spacewalkers wearing Russian Orlan spacesuits.

Tranquility

The Tranquility node was connected to the Earth-facing side of the Unity node in February 2010. The pressurized section provides additional room for crew members and many of the space station's life support and environmental control systems already on board. These systems include air revitalisation, oxygen generation and water recycling. A waste and hygiene compartment and a treadmill also will be relocated from other areas of the station. It also provides an additional docking point for space shuttles and other crew vehicles visiting the station. Attached to the node will be Cupola, a unique work module with six windows on the sides and one on top.

Rassvet

The Russian module Rassvet – which translates as “dawn” – was attached to the ISS in May 2010, where it provides an additional docking port for Russian Soyuz and Progress vehicles. Weighing more than 5000 kilograms and measuring 6 metres in length and 2.3 metres in diameter, Rassvet will host a wide variety of biotechnology and biological science experiments and fluid physics and educational research. The module contains a pressurised compartment with eight workstations equipped with facilities such as a glovebox to keep experiments separated from the in-cabin environment; two incubators to accommodate high- and low-temperature experiments; and a custom platform that protects payloads and experiments from onboard vibrations. The module contains four other workstations, complete with mechanical adapters, to install payloads into roll-out racks and shelves.

Leonardo

When the ISS was first conceived, the Italian Space Agency was assigned the task of building Multi-Purpose Logistics Modules – pressurised modules that would be used to ferry cargo to and from the ISS aboard shuttle flights. In early 2011, the one of the three MPLMs, Leonardo, was refitted, redesigned and permanently attached to the ISS. The cylindrical module is approximately 6.5 metres long and 4.5 metres in diameter. It can carry up to 10 tons of cargo packed into 16 standard Space Station equipment racks. Of the 16 racks the module can carry, five can be furnished with power, data and fluid to support a refrigerator freezer. The extra space it provides will give the station's resident crew what amounts to a giant